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6.4.5 Is LU with Partial Pivoting Stable?

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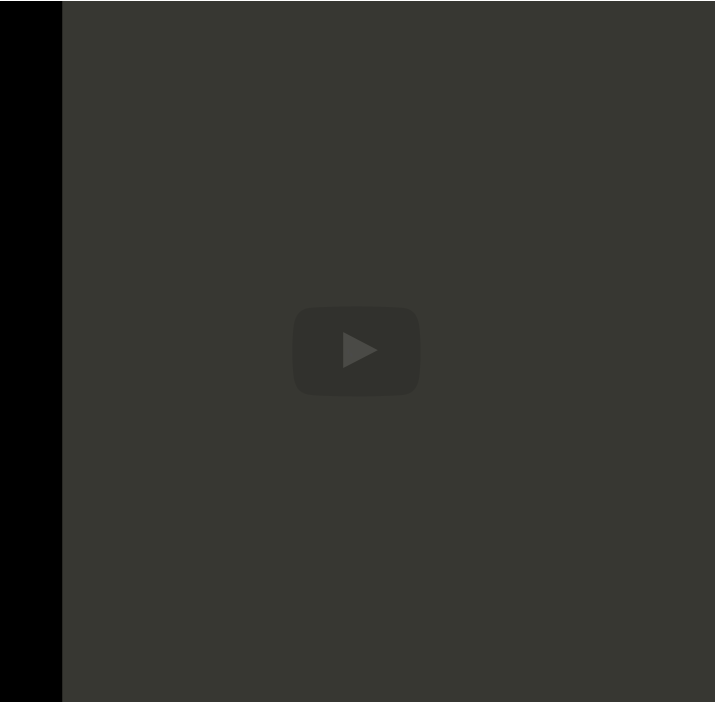
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differ from the original matrix, we're going to have to do an exercise that shows that actually you could see pretty dramatic element growth in U. In other words, the elements in U actually could get pretty large even when you do LU with pivoting. Now, not to worry. It has been observed that in practice, practical problems don't lead to this kind of element growth. And therefore LU with partial pivoting used to solve linear

Video

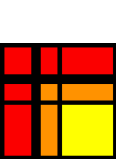
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Reading assignment

1/1 point (graded)



Read Unit 6.4.5 of the notes. [\[LINK\]](#)

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Homework 6.4.5.1

1/1 point (graded)

Apply LU with partial pivoting to $A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & -1 & 1 \end{pmatrix}$.

Give the matrix U that results when performing LU factorization with row (partial) pivoting. Pivot only when necessary.

1	0	1
✓ Answer: 1	✓ Answer: 0	✓ Ar
0	1	2
✓ Answer: 0	✓ Answer: 1	✓ Ar
0	0	4
✓ Answer: 0	✓ Answer: 0	✓ Ar

Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 6.4.5.2

1/1 point (graded)

Generalize the insights from the last homework to a $n \times n$ matrix.

☒ done

✓

What is the maximal element growth that is observed?

☐ $\approx 2n$

☐ $\approx 2^n$

☒ $\approx 2^{n-1}$

✓

Solution:

For a complete solution, see [the notes](#).

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